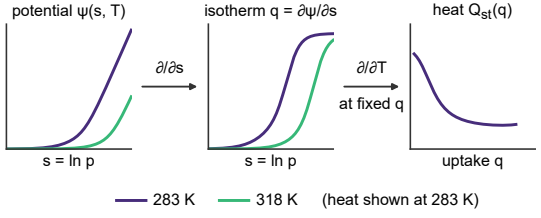
a One potential gives the isotherm and the heat (curves illustrative)



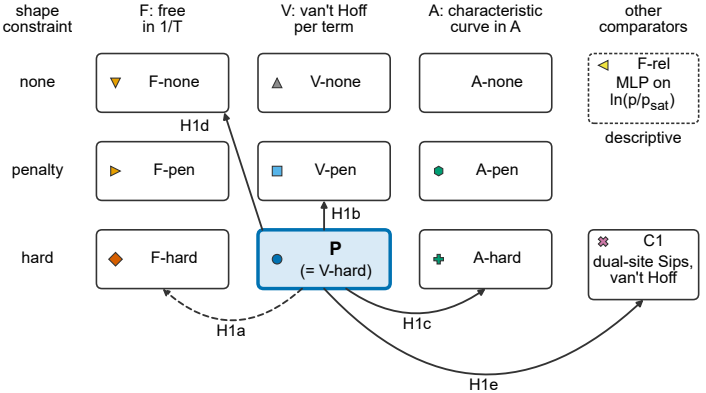
Holds by construction, for every parameter value

- monotone in p: $\partial q/\partial s = \partial^2\psi/\partial s^2 > 0$, because ψ is convex in s
- Henry limit: q grows in proportion to p as $p \to 0$ (Langmuir terms)
- heat always finite: $Q_{st} = -RT^2\,(\partial q/\partial T)\,/\,(\partial q/\partial s)$, denominator never zero; sign and size not constrained

In P each term has a van't Hoff affinity $K_0 \exp(Q / RT)$, Q free in sign.

Every model's heat is computed from its own isotherm by the same code.

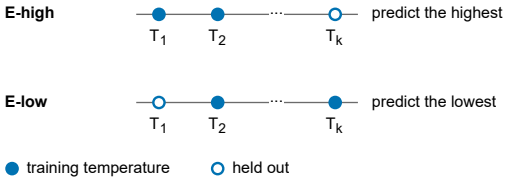
b Factorial: shape constraint × temperature structure



Comparisons of P (H1b primary; H1b-H1e Holm-adjusted; H1a descriptive):

H1a against F-hard: temperature structure
H1b against V-pen: hard against penalty
H1c against A-hard: characteristic curve
H1d against F-none: generic network
H1e against C1: classical dual-site Sips

c Tasks and scores (per material, temperatures $T_1 < \ldots < T_k$)



scored

Accuracy (H1, primary)
E = RMSE / q_{ref} on the held-out isotherm; per material, the mean of $\log_{10}$ E over E-high and E-low

Violations (H3)
grid to ±15 K, $p \le 0.9\ p_{sat}$: uptake falling by more than 0.02 q_{ref} as p rises, or below −0.02 q_{ref} (band to ±20 K reported apart)

Heat (H2, synthetic)
Q_{st} against the exact heat of five synthetic truths at matched loading in 0.1 to 0.9 q_{ref}, on the same tasks